

IBRI, Oman

WMO: 412520

Lat: 23.1954N

Lon: 56.4295E

Elev: 1073

StdP: 14.14

Time Zone: 4.00 (E04)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 52.5	(c) 55.1	(d) 17.9	(e) 14.1	(f) 83.1	(g) 21.7	(h) 16.9	(i) 83.4	(j) 24.4	(k) 70.6	(l) 22.7	(m) 73.0	(n) 6.4	(o) 310	(p) 0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	114.9	69.6	113.1	69.8	111.4	69.9	77.9	97.0	76.8	97.6	75.9	97.2	11.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	127.1	87.2	71.3	120.0	87.7	69.8	114.1	87.9	42.3	97.8	41.2	97.1	40.3	97.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.5	17.8	47.5	118.2	2.6	1.0	45.6	119.0	44.1	119.6	42.6	120.2	40.7	120.9
			41.7	80.3	2.6	2.0	39.8	81.7	38.4	82.9	36.9	84.0	35.1	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.7	67.9	71.7	78.7	86.6	94.8	98.5	100.2	98.3	94.5	87.9	77.9	70.8
	DBStd	12.15	4.03	6.04	5.42	5.59	4.00	3.72	3.43	4.00	3.56	3.48	4.00	3.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	33	18	11	1	0	0	0	0	0	0	0	0	3
	CDD50	13040	555	606	890	1099	1390	1456	1555	1497	1336	1175	837	644
	CDD65	7599	108	197	426	649	925	1006	1090	1032	886	710	387	183
	CDH74	118164	712	1787	5034	9453	15560	17636	19446	17967	14883	10491	3895	1300
CDH80	80919	99	587	2479	5841	11185	13338	14983	13517	10594	6398	1602	296	
Wind	WSAvg	8.0	7.7	8.8	8.7	9.1	8.9	9.2	8.4	7.5	7.1	6.4	6.4	7.4
Precipitation	PrecAvg	4.8	0.6	0.7	1.0	0.6	0.2	0.1	0.3	0.2	0.2	0.2	0.2	0.5
	PrecMax	11.2	1.7	3.1	6.1	3.0	0.6	0.6	1.9	0.7	0.7	0.9	1.3	2.3
	PrecMin	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.4	0.6	1.2	0.7	0.2	0.2	0.4	0.2	0.1	0.2	0.3	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.0	92.9	99.4	105.7	115.0	116.8	116.8	115.0	112.0	105.3	94.8	90.5
		MCWB	59.4	62.1	62.1	66.0	67.3	68.9	70.3	70.8	69.3	66.3	64.0	62.2
	2%	DB	82.6	89.6	96.3	103.5	110.7	114.8	114.8	112.8	109.3	102.9	92.2	85.4
		MCWB	59.6	61.7	61.7	65.3	67.5	68.6	70.5	71.3	69.2	65.9	64.2	60.4
	5%	DB	80.3	86.7	93.7	101.3	108.3	112.9	113.0	111.1	107.4	100.8	90.3	83.1
		MCWB	59.3	60.8	62.0	64.6	67.3	68.4	70.7	71.3	69.0	65.2	63.7	60.8
	10%	DB	78.1	83.6	91.3	99.2	106.1	110.7	111.2	109.2	105.5	98.4	88.2	80.8
		MCWB	58.6	60.2	61.6	63.8	67.0	68.5	70.6	71.3	69.0	65.0	63.1	60.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.8	68.5	70.4	72.9	76.1	78.7	80.4	79.4	76.7	74.2	70.2	67.5
		MCDB	76.6	80.6	82.2	84.8	92.6	92.5	97.3	98.4	95.1	88.1	82.5	77.2
	2%	WB	63.3	66.1	68.3	70.9	74.4	77.1	78.2	77.1	75.4	72.6	68.4	65.7
		MCDB	75.9	80.3	81.7	84.2	92.9	95.3	97.9	99.4	94.5	86.5	82.8	76.7
	5%	WB	61.7	64.5	66.7	69.2	72.7	75.9	77.0	76.2	74.4	71.0	67.2	64.1
		MCDB	75.2	79.4	82.9	87.0	93.5	95.8	98.2	98.6	94.1	87.9	82.2	75.8
	10%	WB	60.2	63.0	65.2	67.5	70.9	74.4	76.0	75.4	73.5	69.3	65.9	62.8
		MCDB	74.3	79.0	82.8	90.7	95.7	96.4	98.1	97.8	93.6	89.3	81.3	76.1
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.5	22.8	21.8	22.3	23.8	21.8	21.8	22.1	20.9	19.3	20.1
		MCDBR	22.4	22.0	24.1	22.4	24.2	25.1	23.3	22.2	22.8	22.3	20.3	22.8
		MCWBR	8.9	8.4	8.6	7.0	7.6	7.8	6.9	6.2	6.7	7.5	7.6	8.3
	5% WB	MCDBR	19.5	18.2	18.6	18.6	20.7	23.6	21.4	21.7	22.4	20.0	19.0	18.4
		MCWBR	8.8	8.3	8.3	7.4	8.5	8.1	6.1	5.8	6.0	6.9	8.0	7.7
Clear-Sky Solar Irradiance	taub	0.397	0.445	0.506	0.562	0.584	0.661	0.748	0.708	0.594	0.470	0.415	0.392	
	taud	2.179	2.015	1.847	1.752	1.708	1.587	1.487	1.546	1.721	1.977	2.144	2.190	
	Ebn at Noon	271	265	254	243	236	217	199	207	228	254	263	267	
	Edh at Noon	42	53	66	74	77	86	95	89	74	55	43	40	
All-Sky Solar Radiation	RadAvg	1490	1739	1993	2195	2407	2389	2198	2169	2055	1871	1569	1415	
	RadStd	66	77	107	130	60	80	82	53	41	49	30	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.54	+0.94	N/A	N/A	+0.47	+0.52	N/A	N/A	N/A	N/A

Nomenclature: See separate page